Title (0:15). Total: about 23 minutes. This talk presents the groundwork of a paper: a bibliometric and text-as-data study of everything written on central bank digital currencies.

The paper in one slide (1:00).

- One question: why do people write about CBDCs, and how has that changed?
- Five sub-questions structure the talk: what, who, why, how, when.
- Three contributions: the data, validated measures, and the timing of attention.
- Headline: 48,188 texts collected, 17,709 keep CBDC as a theme.

Related work (0:50).

- The topic is not new for bibliometrics: about 77 reviews, most of them since 2023.
- The bibliometric ones rely on Scopus or Web of Science: a few hundred journal articles, often English only (Prodan et al. drop non-English texts).
- The closest to us is Hoang et al. (2023, RIBAF): text mining, but still academic only.
- Our angle: every type of text, every language, and measures that are validated.

Outline (0:15). Method first (collection, scoring), then results (corpus, maps, why and when).

Pipeline (0:45).

- Four families of sources, each harvested into its own raw file.
- Then full text is downloaded and parsed, then everything is merged into one CSV.
- Search vocabulary includes older names (digital fiat currency, Fedcoin) to reach the pre-2017 literature.
- Design choice: collect broadly, decide relevance later.

Press harvesting (0:45).

- Problem: a news search returns at most 100 results, but a busy month has thousands.
- Solution: if a month is full, cut it in two, and repeat until each window holds fewer than the cap, down to one day.
- A month needs at most 5 levels of splitting; the cost is $2L - 1$ requests.

Homonyms (0:45).

- Each concept is a regular expression; the same list drives searching and scoring.
- "CBDC" also means *chronic bullous dermatosis of childhood*: an ambiguous term only counts if the text also talks about money or payments.

Theme score (0:45).

- Not “does the word appear” but “is it the subject”.
- Title: strongest signal. Abstract: share of sentences. Full text: density and spread across the document.
- The max rewards one clear signal; the mean rewards consistency. Keywords weigh less: partly machine-assigned.

Validating the filter (0:50).

- A paper needs to show the filter works: I read a stratified sample without seeing the label.
- Main theme: almost always right. Secondary theme: only a minority really is about CBDCs, so every result is re-run on main theme only.
- Recall loss comes mainly from press items whose text could not be fetched.

Sources (0:35).

- Press articles form the largest group; speeches and official documents are few but long and fully available.
- Completeness: most texts are dated to the day; full text and references are the scarcest fields.

Growth (0:40).

- From 189 in 2019 to 3,167 in 2025.
- Turning point 2020: COVID and the decline of cash, Libra (2019), the ECB report on a digital euro (Oct. 2020).
- Caution: the press bar for 2026 is mostly a harvesting effect (news search returns recent articles first); in the paper, press is normalised by its coverage.

Languages (0:25). English dominates, but Russian is a large second block: a reminder that English-only databases miss part of the debate.

Academic literature (0:45).

- Much of the debate happens in working papers, often by central bank economists.
- The most cited papers are macro-modelling papers from 2016–2020.
- Laws: output grows 58% a year; most authors write once (a Lotka exponent above 2): a young, open field around a small core of sources.

Voices (0:35).

- European Central Bank leads the speeches: the digital euro project makes it the most vocal institution.
- The press is led by specialised fintech outlets before generalist media.

Co-citation (0:50).

- Method in one line: two items are linked when they appear in the same paper, weighted by association strength, grouped by the Louvain algorithm (details in backup).
- Two references are linked when the same paper cites both: this shows what the field builds on.
- Core: Andolfatto (2020), Agur (2019), Bordo (2017).
- Clusters read as schools: CBDC design and bank intermediation models; central bank policy reports; adoption studies.

Co-authorship (0:45).

- Small, separate teams: the field is fragmented, around institutions (BIS, IMF, academic groups).
- Productivity is concentrated on a few very prolific authors.
- Who sets the agenda: central bank and international-institution economists write few of the papers but collect a third of the citations, though their share of papers is shrinking as the field opens up.

Thematic map (0:50).

- Centrality: how connected a theme is to the others. Density: how developed it is internally.
- Motor themes (central and developed): digital transformation in financial services.
- Niche themes are developed but isolated; basic themes are transversal but less structured.

Evolution (0:35).

- Rising fastest: financial inclusion; falling most: payment systems (shares, while the field itself grows).
- The literature moves from "how would it work" to "who adopts it and for whom".
- Bottom line, TF-IDF answers "when": each period's vocabulary compared to the others.

Toolbox (0:45).

- One row per question; the middle column is what the results use.
- The right column matters for the paper: each alternative was run against the same hand-coded texts, and lost.
- Remaining improvements are tuning (more labels, bigger language models), not new methods.

Topics (1:00).

- Unlike keywords, embeddings group texts by meaning and across languages: a Russian and an English paper on the digital ruble land together.
- Two kinds of topics: issues (privacy, cross-border, stablecoins, adoption) and jurisdictions (digital euro, ruble, e-CNY, rupee).
- Jurisdiction topics keep a stable share, but move away from the digital euro towards the ruble, the e-CNY and the rupee: the literature follows national projects.
- Central banks dominate one topic, the digital euro; researchers the rest.

Framing (0:40).

- The prior shrinks rare words, so the list is not dominated by noise.
- academics: *research*, *analysis*, *legal*; central bankers: *payments*, *area*, *europe*; press: *euro*, *rupee*, *yuan*.

Motivations: method (0:35).

- The "why" requires reading, so the reading is done once to build the codebook, then applied to every text.
- A text can have several motivations. Press is excluded: headlines are too short.
- Validation: codebook tuned on 100 texts, then tested on 50 others never used for tuning. High precision, modest recall: what is found is right, but about half is missed (non-English texts, paraphrases). Details in the backup slide.

Motivations: results (1:00).

- The adjusted columns compare like with like: same mix of actors and languages in each period.
- Early literature: monetary policy, how a CBDC changes rates and bank funding. Recent literature: inclusion, cross-border payments, geopolitics, privacy.
- Central bankers and academics do not give the same reasons.

Measuring motivations (0:40).

- Five machine coders against the same hand-coded texts: SBERT in two ways, an NLI model, two local LLMs.
- None beats the codebook; the codebook's weakness is recall, so shares are lower bounds.
- A larger LLM behind an API could do better; it would be checked the same way. That is tuning, not a new method.

Tone (0:50).

- Only sentences that name a CBDC are scored, so this is the tone about CBDCs, not about the economy in general.
- Central banks are the most positive, the press the least.
- Academic tone rises (0.08 in 2019 to 0.24 in 2025), in line with the move from risks (stability) to benefits (inclusion).
- Validation: FinBERT rarely confuses positive with negative, but sees tone in neutral sentences: compare groups and years, do not read levels.

Stance (0:50).

- Tone says how positive the words are; stance says whether the author is for or against CBDCs.
- Levels come from reading: central banks are the advocates, researchers mostly neutral, criticism is rare and mostly in the press.
- Changes come from the language model, run on a random sample: central banks have become much more favourable since 2020, as their own projects advanced.

Bursts (0:45).

- Where the event study tests dates I chose, burst detection lets the data say when a theme suddenly takes off.
- Early bursts: monetary policy and economic impact, economic theories and models, monetary policy. Latest bursts: monetary sovereignty, unified payments interface, systemic risk, artificial intelligence.
- Read with the topics: the field moves from monetary policy to sovereignty, tokenisation and national projects.

Event study (0:50).

- For each event, compare the texts in the two weeks after it with the rate of the three months before.
- The naive Poisson test is far too lenient: a quarter of random dates would look significant. So each event is compared with the placebo dates instead.
- Red lines: $p < 0.05$ against the placebo distribution.
- 6 of 12 events beat 95% of placebo dates; strongest: EU digital euro bill, ECB digital euro report.
- Attention follows institutional milestones (central bank reports, legislation) more than market news. Sand Dollar comes 18 days after the ECB report: the two windows overlap.

Robustness (0:50).

- Each row re-runs the headline results on a different sample: only texts clearly about CBDCs, only English, only journal articles, without the partial year, or with the high-recall coding.
- The direction of every result is unchanged; magnitudes move a little. This is the table that answers a referee's "what if".

Takeaways (0:50).

- Recap the five answers: what, who, why, how, when.
- Main message: CBDC research moved from a monetary-policy question to one of inclusion, cross-border payments and national projects; the tone has become more positive; attention follows central bank milestones.
- Limits are known and measured: each one has a fix in the paper plan.

Paper plan (1:00).

- Data, methods, results and robustness are done; the method for each question was chosen against the alternatives.
- What is left: finishing the press harvest, a second human reader, and writing.
- Thank you; questions.

Backup (only if asked). The reference coding was done by reading each text against the codebook definitions (AI-assisted, spot-checked). Payment efficiency is the weakest category: it is expressed in too many ways for keywords.

Backup (only if asked).

- Association strength corrects for popularity: two very frequent items co-occur often by chance.
- Louvain groups nodes so that links are dense inside groups and sparse between them; Q measures how clear the grouping is.

Backup (only if asked). A speech with one paragraph on the digital euro is "mentioned only"; a medical case report on CBDC (the skin disease) scores zero.

Backup (only if asked).

- APIs return results page by page; a cursor points to the next page until none is left.
- Free APIs limit the request rate: a throttle guarantees a minimum gap Δ between calls, and the program waits longer after each refusal.

Backup (only if asked).

- The same paper appears in several databases: we match on DOI, else on normalised title and year.
- When merging, keep the best of each copy: the most precise date, the longest abstract.

Backup (only if asked). Errors are almost all "toned vs. neutral": FinBERT was trained on financial news, where descriptive sentences about a new product read as positive.

References.